Assignment 5

Due Wednesday 18 March at the beginning of class

This Assignment is based on Chapter 3 of our textbook,[1] especially sections 3.3, 3.4, and 3.6, and on the lectures. However, please also read sections 5.1 and 5.2; this will be the material which we cover next.

DO THE FOLLOWING Exercises from Chapter 3 (see pages 69–73):

- **Exercise 3.3.4**
- **Exercise 3.4.1**
- **Exercise 3.4.2**
- **Exercise 3.6.4** *This problem assumes that p and q are Hölder conjugate exponents: $\frac{1}{p} + \frac{1}{q} = 1$. (Hint. Rewrite this simple definition in some different ways.) Also, I wrote my proof in terms of a single function $\phi = f + g$. If you do this, justify it.*
- **Exercise 3.6.7**

DO THE FOLLOWING ADDITIONAL PROBLEMS.

P7. *The following problem asks for the heart of the correct proof of Theorem 3.22, in the $p = 1$ and $\mathbb{R}^1$ case. The book's definition of step functions is wrong, but the Theorem is true.*

A function $h : X \to \mathbb{R}$ is a *step function* if it is a finite linear combination of characteristic functions of intervals; there are coefficients $c_k \in \mathbb{R}$ and intervals $I_k \subset \mathbb{R}$ so that

$$h(x) = \sum_{k=1}^{n} c_k \mathbb{1}_{I_k}(x).$$

Suppose $E \subset [0,1]$ is measurable. Show that the indicator function $f(x) = \mathbb{1}_E(x)$ can be arbitrarily approximated in L^1 norm by a step function. That is, show that for all $\epsilon > 0$ there exists a step function $g(x)$ so that

$$\|\mathbb{1}_E - g\|_1 = \int_{[0,1]} |\mathbb{1}_E(x) - g(x)|\, dm < \epsilon.$$

(*Hint. Return to the definition of $\mathcal{M}_{\mathcal{F}}$. Use it. The function g is also an indicator.*)

P8. *(Simplified.)* Prove that the triple $(X, \mathcal{R}, \mu)$ given in Example 6 at the beginning of Section 3.6 (page 58) is, in fact, a measure space.

P9. *(Simplified: part (b) removed.)* Suppose that $(X, \mathcal{R}, \mu)$ is a measure space. We say that two measurable functions $f, g : X \to \mathbb{R}$ are *equal almost everywhere* if the set of points where they differ has μ-measure zero. Prove that this is an equivalence relation on the set of such measurable functions.

[1] K. Saxe, *Beginning Functional Analysis*, Springer 2010.

P10. Suppose that $(X, \mathcal{R}, \mu)$ is a *finite* measure space, that is, suppose that $X \in \mathcal{R}$ and $\mu(X) < \infty$. For $f : X \to \mathbb{C}$ measurable (with respect to $\mathcal{R}$) and $1 \leq p < \infty$, define $L^p(X, \mu)$ and $\|f\|_p$ as usual; see page 59. Consider two different powers, $1 \leq p < q < \infty$. Show that $L^q(X, \mu) \subset L^p(X, \mu)$. Give an example of a function on $(X, \mu) = ([0, 1], m)$, which will depend on p and q, for which this is a proper inclusion.

P11. *The requested proofs will help you appreciate Hölder's and Minkowski's inequalities. These results are used in Theorems 3.17, 3.18, and 3.19. For part* **(a)**, *start with the triangle inequality on* $\mathbb{R}$, *and also* $|u| + |v| \leq 2\max(|u|, |v|)$. *A hint on part* **(b)** *is on page 60.*

(a) Suppose $1 \leq p < \infty$ and let $u, v \in \mathbb{R}$. Show that

$$|u + v|^p \leq 2^p \left(|u|^p + |v|^p\right).$$

(b) Suppose $1 \leq p, q < \infty$ are (Hölder) conjugate exponents, $\frac{1}{p} + \frac{1}{q} = 1$. Also suppose $x, y \geq 0$. Prove *Young's inequality*:

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q}.$$

Extra Credit. Suppose $1 \leq p < q < \infty$, and consider ℓ^p and ℓ^q. Prove that for $f \in \ell^1$,

$$\|f\|_\infty \leq \|f\|_q \leq \|f\|_p \leq \|f\|_1.$$

Also prove that $\ell^p \subset \ell^q$—this inclusion reverses the direction from **P10**—and give an example to show that this is a proper inclusion.